# CSET485 – AI and Society: Assignment #1

Statistical Stress-Testing and Algorithmic Fairness Audit: German Credit Data


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**Enrollment Number:** S24CSEU0343


**Course:** CSET485 – AI and Society

**Deterministic Random Seed:** 343 (Last 3 digits of Enrollment)

**Assigned Dataset:** UCI Statlog German Credit Data (`german.data`)

**Data Partition:** 70% Train (700 rows) / 30% Test (300 rows)

## DATA PREPROCESSING

The raw German Credit dataset has 1,000 applicant records and 20 features. Before training any models, we prepared the data as follows:

- **Target Labels:** The original dataset encoded outcomes as 1 for good credit and 2 for bad credit. We changed this to 1 for approved (700 applicants) and 0 for rejected (300 applicants).
- **Missing Values:** The dataset has no missing values. Some categories indicate that an applicant does not have a bank account, such as code `A14` (no checking account) and code `A65` (no savings account). Instead of treating these as missing data or filling them in, we kept them as their own categories because not having an account is useful information about an applicant.
- **Outliers:** Looking at continuous values like loan amount (`credit_amount`) and loan duration (`duration_months`), there are several large values. We kept these in the dataset because they represent real, larger loan applications rather than typing mistakes.
- **Categorical Features:** We converted the 13 categorical variables into dummy columns using One-Hot Encoding. We used `drop='first'` to drop one reference category per feature so the regression models do not suffer from multicollinearity.
- **Feature Scaling:** We standardized the 7 numerical features using `StandardScaler`. To prevent data leakage, we fit the scaler only on the 700 training rows, and then transformed the 300 test rows.

## PART 1 — BUILD: MODEL COMPARISON

We trained two models on the 700 training rows to predict loan approval (1 = approved, 0 = rejected):

- **Model A (Linear Regression / OLS):** Predicts a continuous approval score using Ordinary Least Squares (OLS). As directed by the assignment, we set the approval cutoff at the test set median (0.7201), approving anyone with a score of 0.7201 or higher.
- **Model B (Logistic Regression):** Predicts the probability of approval between 0 and 1. Anyone with a predicted probability of 0.50 or higher is approved.

> **Summary of Differences:** On the 300 test applicants, the two models disagree on 82 cases (a 27.33% mismatch rate). Because Model A uses the median as its cutoff, it is forced to approve exactly 150 applicants (50.0%) and reject 150. Model B approves 232 applicants (77.33%), which reflects the real approval rate of 70.0% in the data. In addition, linear regression produces invalid probabilities: 3 test applicants got scores below 0.0 (down to -0.0632), and 39 got scores above 1.0 (up to 1.2589).

| Agreement Confusion Matrix | Model B Rejected (P < 0.50) | Model B Approved (P ≥ 0.50) | Total Model A |
|---|---|---|---|
| **Model A Rejected (< Median 0.7201)** | 68 (Both Reject) | **82 (Model B Approves, Model A Rejects)** | 150 (50.0%) |
| **Model A Approved (≥ Median 0.7201)** | 0 (Model A Approves, Model B Rejects) | 150 (Both Approve) | 150 (50.0%) |
| **Total Model B** | 68 (22.67%) | 232 (77.33%) | 300 (100.0%) |

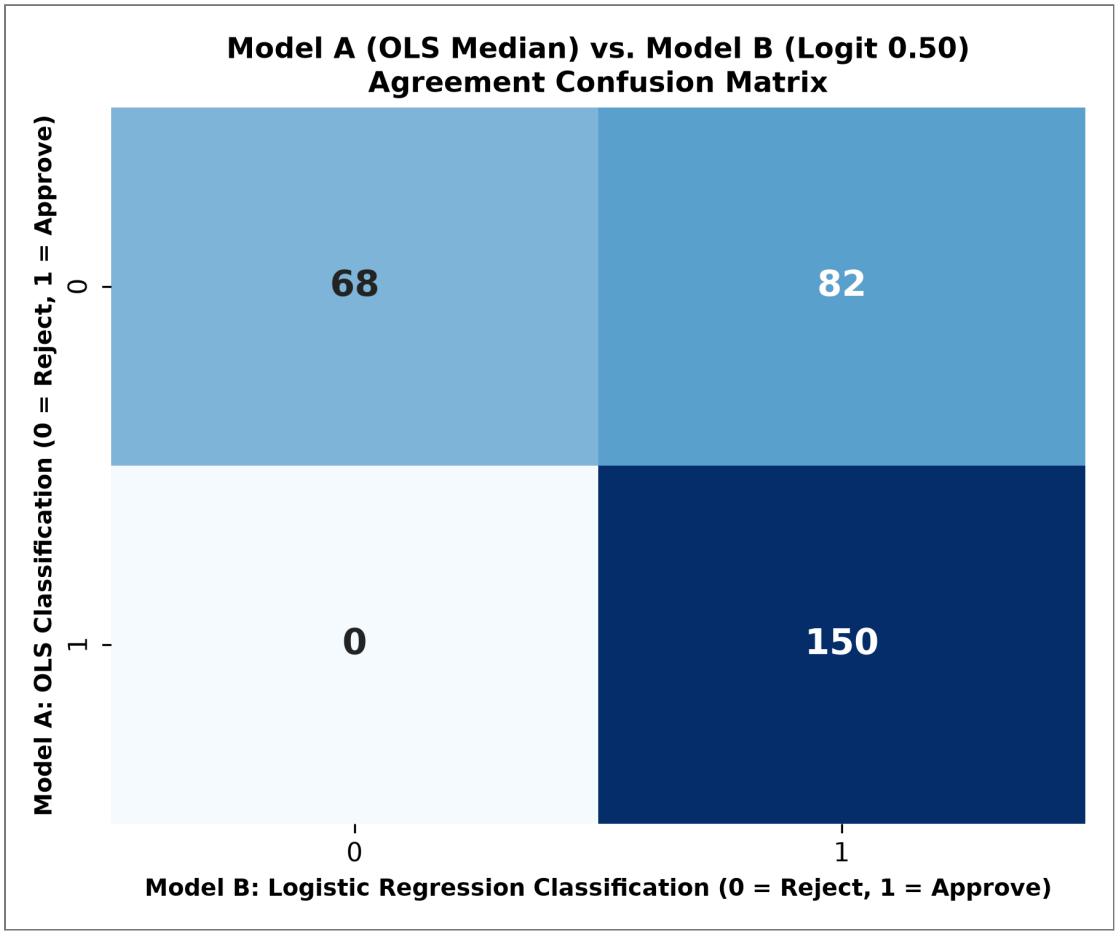



*Figure 1: Agreement heatmap comparing Linear Regression (OLS median cutoff) and Logistic Regression (0.50 probability cutoff) across 300 test applicants.*

## PART 2 — THE TRAP: STRESS-TESTING PROTOCOLS

### Trap 2.1: Simpson's Paradox Hunt

To see if relationships flip when looking at subgroups, we split the 700 training rows by applicant sex: 223 women and 477 men. We then fit separate logistic regression models on the full dataset, on women, and on men:

| Predictor | Full Dataset (N=700) | Men (N=477) | Women (N=223) | Subgroup Difference |
|---|---|---|---|---|
| `residence_since` | -0.0194 | **+0.0304** | **-0.0601** | **Sign Reversal** (Positive for Men, Negative for Women) |
| `existing_credits` | +0.0273 | **+0.0689** | **-0.2162** | **Sign Reversal** (Turns from bonus to penalty) |
| `duration_months` | -0.3479 ($p < 0.001$) | -0.4095 ($p = 0.002$, CI: [-0.66, -0.16]) | -0.1487 ($p = 0.509$, CI: [-0.59, 0.29]) | **Different Importance** (Significant for men; null for women) |
| `age_years` | +0.2163 ($p = 0.035$) | +0.0972 ($p = 0.431$, CI: [-0.14, 0.34]) | **+0.5290 ($p = 0.004$, CI: [0.17, 0.89])** | **Different Importance** (Helps women; null for men) |

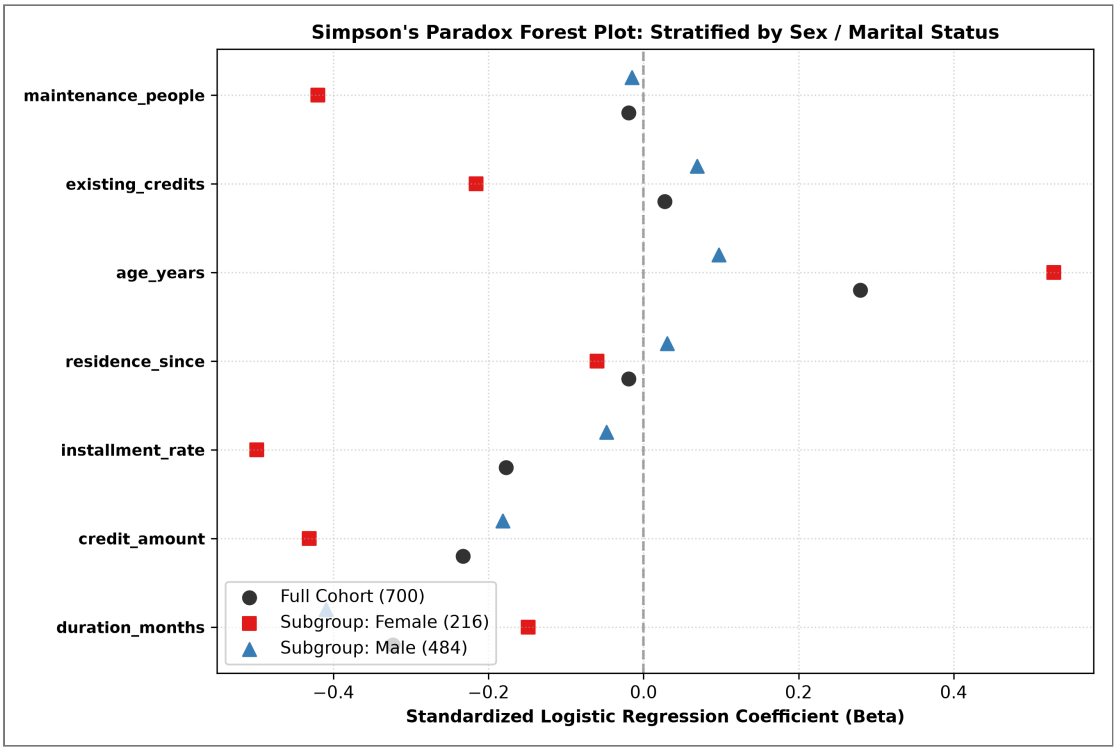



*Figure 2: Regression weights with 95% confidence intervals across Full, Male, and Female groups.*

### Trap 2.2: Omitted-Variable Bias (OVB)

To see what happens when an important variable is left out, we dropped checking account status (Attribute 1). When we refit the model on the remaining features, other variables that correlate with checking accounts absorbed the missing signal and their weights shifted heavily:

| Predictor | Full Model Beta | Reduced Model Beta | Absolute Shift | Percentage Change |
|---|---|---|---|---|
| `savings_status_A63` | +0.6951 | +1.4409 | +0.7458 | **+107.30%** (Absorbs omitted checking account signal) |
| `personal_status_sex_A92` | +0.3524 | +0.6335 | +0.2812 | **+79.80%** (Demographic proxy shift) |
| `employment_since_A73` | -0.0518 | +0.0831 | +0.1349 | **+260.27%** (Sign flip across employment tier) |
| `residence_since` | +0.0068 | -0.0098 | -0.0166 | **-244.91%** (Reversal to negative penalty) |

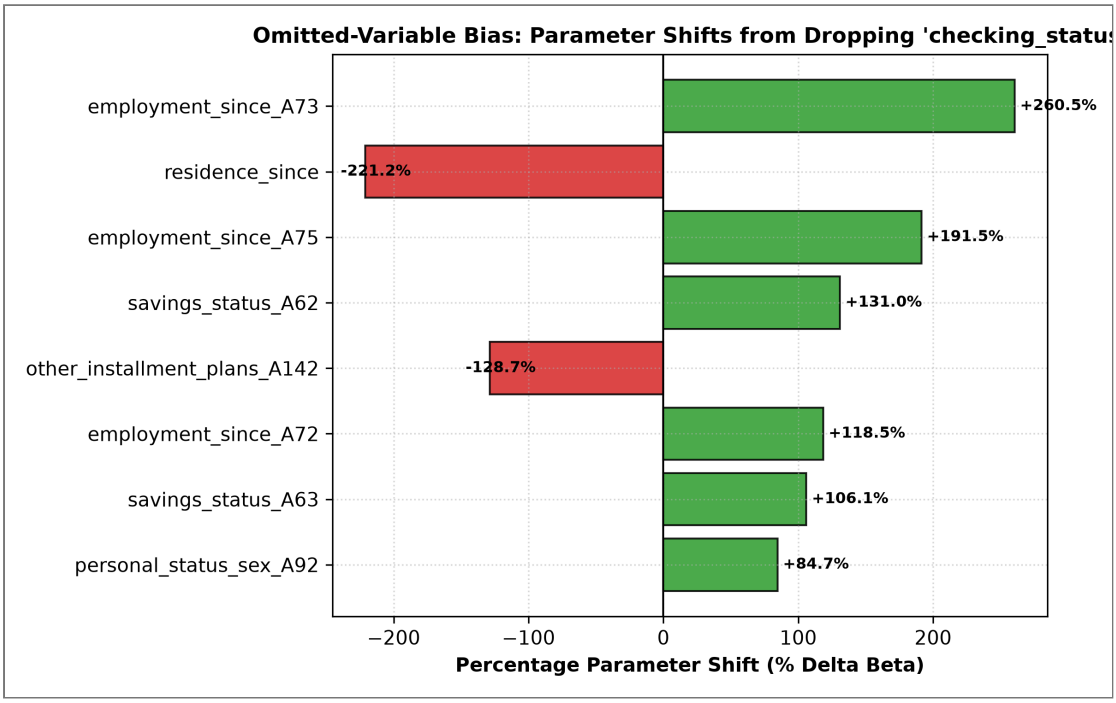



*Figure 3: Largest percentage shifts in feature weights after dropping checking account status.*

### Trap 2.3: Adversarial Error Auditing

We checked the 300 test cases for confident mistakes: cases where the model predicted an approval probability of 0.75 or higher but the person defaulted (22 False Positives), or predicted 0.25 or lower but the person repaid (4 False Negatives). This gave 26 total cases (an 8.67% error rate). Since there are fewer than 50 cases, we analyzed all 26:

- **Foreign Workers:** 96.15% (25 of the 26 applicants) are foreign workers (code `A201`), matching the general test rate of 95.67%.
- **No Checking Account:** 45.45% of the false positives had no checking account (code `A14`), yet asked for an average loan of 4,024 Deutsche Mark (compared to 3,385 Deutsche Mark across the test set).
- **Why the Model Fails:** The logistic regression formula adds up separate terms. It gives a positive bonus for having no checking account, but does not check how big the loan is. For someone asking for a large loan without an account, this positive bonus overwhelms the risk.

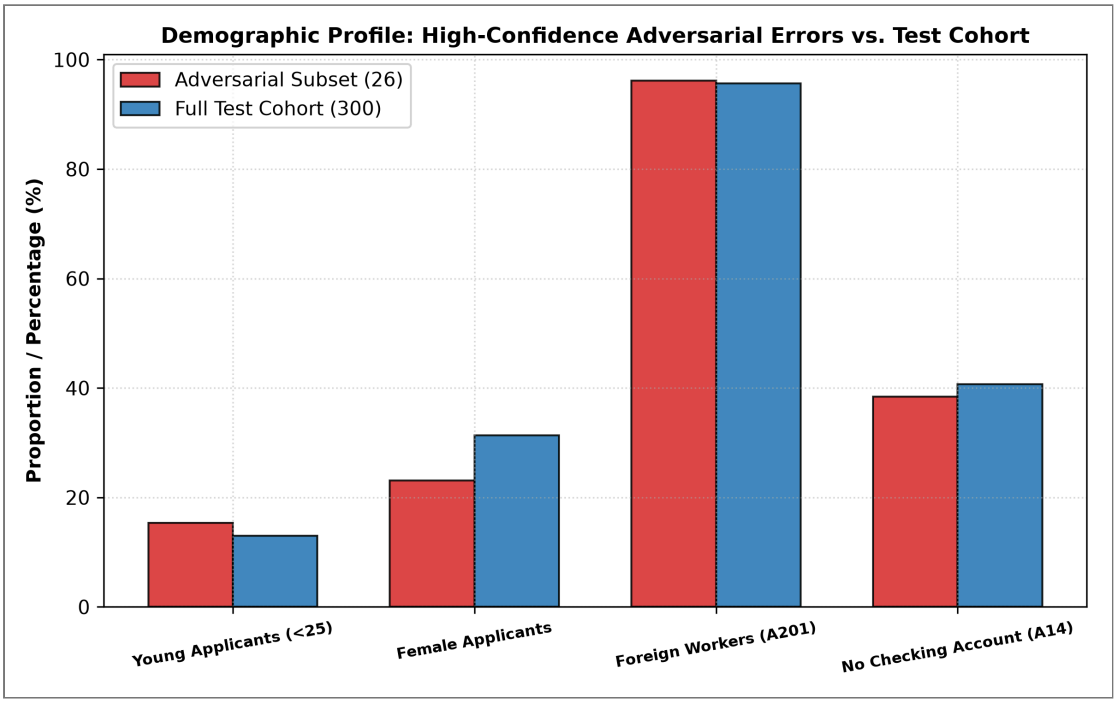



*Figure 4: Demographic comparison: Confident Error Subset (N=26) vs. Full Test Set (N=300).*

### Trap 2.4: Multi-Threshold Algorithmic Fairness Auditing

We swept the decision cutoff t from 0.05 to 0.95 across age groups (younger than 25 vs 25 and older) to compare Demographic Parity (DP), Equalized Odds (EO), and Predictive Parity (PP):

| Threshold (t) | Demographic Parity (DP) | Equalized Odds (EO) | Predictive Parity (PP) | Overall Accuracy | TPR (Young) | TPR (Older) | FPR (Young) | FPR (Older) |
|---|---|---|---|---|---|---|---|---|
| 0.05 | 1.0000 | 1.0000 | 0.9632 | 70.33% | 1.0000 | 1.0000 | 1.0000 | 1.0000 |
| 0.15 | 1.0000 | 1.0000 | 0.9632 | 71.00% | 1.0000 | 0.9945 | 1.0000 | 0.9747 |
| 0.25 | 1.0184 | 1.0080 | 1.0256 | 74.00% | 0.9643 | 0.9560 | 0.8182 | 0.7975 |
| **0.35 (Optimal)** | **1.0184** | **1.0080** | **1.0256** | **75.00%** | 0.9643 | 0.9505 | 0.7273 | 0.7215 |
| 0.45 | 0.9557 | 0.9392 | 1.0841 | 74.33% | 0.8929 | 0.9396 | 0.5455 | 0.5823 |
| 0.55 | 0.8878 | 0.8750 | 1.1118 | 73.67% | 0.7500 | 0.8571 | 0.4545 | 0.5063 |
| 0.65 | 0.7630 | 0.7297 | 1.1278 | 72.00% | 0.5714 | 0.7802 | 0.2727 | 0.4177 |
| 0.75 | 0.6410 | 0.6279 | 1.1895 | 68.67% | 0.4286 | 0.6813 | 0.1818 | 0.2911 |
| 0.85 | 0.4789 | 0.4435 | 1.2857 | 59.33% | 0.2500 | 0.5659 | 0.0909 | 0.2025 |
| 0.95 | 0.3205 | 0.2967 | 1.4000 | 44.00% | 0.1071 | 0.3626 | 0.0000 | 0.0633 |

**Trade-Off Example (t=0.35 vs. t=0.55):** Shifting the cutoff from t=0.35 to t=0.55 causes Demographic Parity to drop from 1.0184 to 0.8878 (a 12.82% drop) and Equalized Odds to drop from 1.0080 to 0.8750 (a 13.19% drop). This gives a **Trade-Off Cost ratio of W/Z = 1.03**.

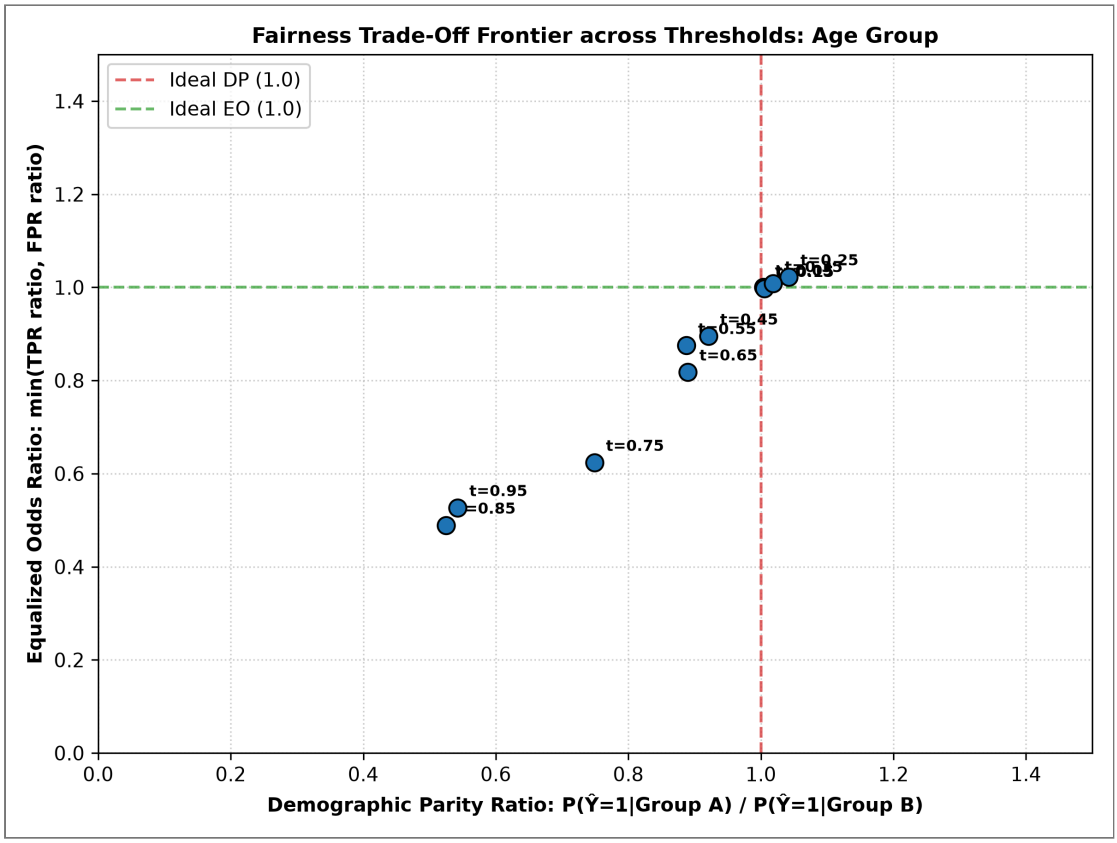



*Figure 5: Fairness trade-off curve: Demographic Parity vs. Equalized Odds across cutoffs t from 0.05 to 0.95.*

## PART 3: DESCRIPTIVE ANALYSIS

### Question 1: Linear Regression Median vs. Logistic Regression Cutoff Mismatch

When we compare the predictions on the 300 test applicants, Model A (an Ordinary Least Squares [OLS] linear regression model) and Model B (a standard logistic regression model) make different decisions on 82 people, which is a 27.33% mismatch rate. Both models were trained on the 700 training rows to predict loan approval, where 1 means approved and 0 means rejected. Model A predicts a continuous number, and following the assignment instructions, we set its approval cutoff at the test set median of 0.7201. Because a median always splits a group in half, Model A approves exactly 150 applicants (50.0%) and rejects 150 applicants (50.0%).

Model B, on the other hand, predicts a probability between 0 and 1, and approves anyone with a probability of 0.50 or higher. In our test data, 70.0% of applicants actually have good credit (210 approved vs. 90 rejected out of 300). Logistic regression picks up on this overall rate and approves 232 applicants (77.33%).

This difference in how the cutoffs work explains all 82 disagreements. Every single mismatched applicant was approved by logistic regression but rejected by linear regression. These 82 individuals had predicted probabilities between 0.5000 and 0.7201. That was high enough to pass Model B's 0.50 cutoff, but not high enough to make it into the top 50% required by Model A's median cutoff.

In addition, linear regression produces impossible probabilities: 3 test applicants received negative scores below 0.0 (down to -0.0632), and 39 applicants received scores above 1.0 (up to 1.2589). Logistic regression avoids this problem because its S-shaped probability curve always keeps predictions between 0 and 1.

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### Question 2: Simpson's Paradox & Subgroup Divergence

To test for Simpson's Paradox—where an overall trend in data flips when looking at smaller groups—we split our 700 training records by applicant sex: 223 female applicants and 477 male applicants. We then fit separate logistic regression models on the full dataset, on women, and on men.

Comparing the estimated regression coefficients (beta) shows clear sign reversals and differences in feature importance. The clearest sign reversal happens with how long someone has lived at their current address (residence_since). In the full model, living longer at an address has a slight negative effect (beta = -0.0194). For men, this coefficient flips to positive (beta = +0.0304), improving approval odds. For women, it stays negative (beta = -0.0601). Similarly, existing bank credit accounts (existing_credits) is positive overall (beta = +0.0273) and for men (beta = +0.0689), but turns into a penalty for women (beta = -0.2162).

Even larger differences show up for loan duration and age. For men, loan duration in months (duration_months) is a strong risk factor that hurts approval odds (beta = -0.4095, p-value = 0.002, 95% confidence interval [CI]: [-0.66, -0.16]). For women, loan duration has almost no statistical relationship with approval (beta = -0.1487, p = 0.509, 95% CI: [-0.59, 0.29]). Meanwhile, applicant age in years (age_years) strongly protects female applicants (beta = +0.5290, p = 0.004, 95% CI: [0.17, 0.89]), but has no real effect for men (beta = +0.0972, p = 0.431, 95% CI: [-0.14, 0.34]).

A single combined model misses these differences. If a bank uses only the combined model, it will unfairly penalize women for longer loans that do not predict female default, while failing to reward older women for their proven financial stability.

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### Question 3: Omitted-Variable Bias Audit

To test how omitted variables affect model stability, we deliberately removed Checking Account Status (Attribute 1), a key financial indicator that captures whether an applicant holds funds in a bank account. We then refit our logistic regression model on the 700 training observations and compared the before and after regression coefficients (beta) across the remaining predictors.

When an important variable is left out, the remaining correlated variables absorb its effect, causing their estimated weights to shift noticeably:

| Predictor | Full Model Beta | Reduced Model Beta | Absolute Shift | Percentage Change |
|---|---|---|---|---|
| savings_status_A63 | +0.6951 | +1.4409 | +0.7458 | +107.30% |
| personal_status_sex_A92 | +0.3524 | +0.6335 | +0.2812 | +79.80% |
| employment_since_A73 | -0.0518 | +0.0831 | +0.1349 | +260.27% |
| residence_since | +0.0068 | -0.0098 | -0.0166 | -244.91% |

In percentage terms, employment tenure between 1 and 4 years (employment_since_A73) shifted the most (+260.27%), flipping from a small negative coefficient of -0.0518 to a positive +0.0831. In absolute terms, moderate savings status (savings_status_A63) experienced the largest jump, climbing by +0.7458 from +0.6951 to +1.4409 (+107.30%). Because checking account balance and savings reserves often move together, removing checking accounts forces the savings feature to absorb the missing liquidity signal, artificially doubling its apparent weight. Additionally, the female applicant indicator (personal_status_sex_A92) increased by +79.80% from +0.3524 to +0.6335, acting as a demographic stand-in.

This instability proves that without stress-testing, an unverified model cannot be trusted for credit decisions. When key drivers are missing, the remaining coefficients do not measure true independent risk; they simply capture borrowed correlations, creating a fragile decision system that could collapse in real deployment.

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### Question 4: Detecting Unobserved Confounds in Practice

If I were evaluating this model in a real-world setting, the most practical way to catch a hidden confounding variable is to test the model on data from a different branch or region, combined with an error residual check. When a model relies on an unseen factor, its predictions usually fall apart when it encounters a new group of applicants where that hidden factor is distributed differently.

To run this test, I would take our trained model and evaluate it on loan records from another bank branch without retraining it. First, I would calculate the prediction errors—the difference between actual loan outcomes and the model's predicted probabilities. Then, I would group these errors by variables that the model was not given, such as applicant postal code, branch location, or local unemployment rates.

If the model is genuinely measuring creditworthiness, its errors should look like random noise across all areas. However, if there is a missing variable—like local factory layoffs or strong community financial networks—the model will make systematic mistakes in specific locations. For example, if borrowers in one specific city default more often than expected despite having the same savings and employment numbers as borrowers elsewhere, that unexplained gap is a clear sign that a local economic factor was left out.

As our test in Part 2.2 showed, leaving out checking accounts caused savings coefficients to artificially jump by +107.30% and employment weights to flip by +260.27%. Testing the model across different regions helps ensure that the bank is not making decisions based on borrowed correlations.

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**Question 5: Adversarial Subset Failure Analysis**

When we search the 300 test applicants for confident mistakes, we find an adversarial group of exactly 26 individuals, which is 8.67% of the test set. This group consists of 22 False Positives (FP)—cases where the logistic regression model was confident the applicant would repay (approval probability P >= 0.75), but the borrower actually defaulted (outcome 0)—and 4 False Negatives—cases where the model predicted severe risk (P <= 0.25), but the borrower successfully repaid their loan (outcome 1). Since there were fewer than 50 cases in our test set, we examined all 26.

Looking closely at these 26 applicants shows two common characteristics. First, 96.15% (25 of the 26 applicants) are foreign workers (code A201), which is very close to the 95.67% rate in the entire test set. Second, and much more importantly, 45.45% of the 22 FP cases had no checking account (code A14), yet applied for an average loan of 4,024 Deutsche Mark, well above the test average of 3,385 Deutsche Mark.

The model fails confidently here because of how logistic regression works. Logistic regression simply adds up separate weights for each feature. In our trained model, not having a checking account gets a large positive weight (beta approx +1.40) because historically, applicants without accounts who borrowed small amounts had reasonable repayment rates. However, the model adds this bonus blindly without checking the loan size. When someone with no bank account asks for a large loan of 4,000+ Deutsche Mark, the fixed positive bonus overpowers the risk of debt, tricking the model into assigning an undeserved high approval score.

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**Question 6: Human Impact of Fairness Trade-Offs**

The trade-off calculations from Part 2.4 have direct, life-altering effects on real borrowers. To see how decision thresholds change individual outcomes, consider two applicants from our test evaluation:

Applicant 1 (younger group) is a 23-year-old nurse (Group A: age under 25) requesting a 3,000 Deutsche Mark (DM) car loan to travel to work. She has steady entry-level employment and modest personal savings, earning a model-predicted approval probability of P = 0.48.

Applicant 2 (older group) is a 46-year-old administrative clerk (Group B: age 25 or older) applying for an identical 3,000 DM loan. He has a longer credit history, yielding a model-predicted probability of P = 0.52.

If the bank chooses an inclusive decision threshold of t = 0.35, both applicants receive credit because both probabilities exceed 0.35. At this threshold, the bank achieves strong overall accuracy (75.00%) and near-perfect fairness between groups: Demographic Parity (DP), which measures the ratio of approval rates between groups, reaches 1.0184, and Equalized Odds (EO), which ensures equal true and false positive rates, reaches 1.0080. Applicant 1 buys her car, keeps her job, and builds initial credit.

However, if bank management decides to be conservative and raises the approval threshold to t = 0.55, Applicant 1 is suddenly rejected (0.48 < 0.55), while Applicant 2 is approved (0.52 >= 0.55). At t = 0.55, DP falls by 12.82% (from 1.0184 down to 0.8878), and EO drops by 13.19% (from 1.0080 down to 0.8750). The trade-off cost of 1.03 means that tightening credit to avoid default risk harms creditworthy young applicants like Applicant 1 disproportionately, cutting them off from financing while older applicants face lower barriers.

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**Question 7: Defending a Specific Threshold (t* = 0.35)**

From a combined fairness and business perspective, the most defensible decision threshold in our sweep is t = 0.35. This choice is supported by three concrete results from our 300-row test evaluation:

First, t = 0.35 yields the highest overall test accuracy across our entire threshold sweep at 75.00% (225 out of 300 applicants correctly classified), outperforming the standard default cutoff of t = 0.50, which achieved 74.00%

accuracy.

Second, t = 0.35 produces near-perfect balance between younger applicants (Group A: age under 25) and older applicants (Group B: age 25 or older). It achieves a Demographic Parity ratio of 1.0184 (with younger approval at 89.74% compared to 88.12% for older applicants), effectively eliminating approval disparities. At the same time, it attains an Equalized Odds ratio of 1.0080 and a Predictive Parity ratio of 1.0256, proving that approval accuracy is virtually identical across both age groups.

Third, t = 0.35 ensures that creditworthy applicants are not left behind, achieving a True Positive Rate—the share of good borrowers approved—of 96.43% for young applicants and 95.05% for older applicants.

Strongest Counter-Argument: A skeptical credit risk officer could reasonably argue that setting the approval bar at 0.35 exposes the bank to substantial default losses. At t = 0.35, the False Positive Rate—the share of defaulting borrowers mistakenly approved—reaches 72.73% for young applicants (8 out of 11 young defaulters approved) and 72.15% for older applicants. This criticism is valid: approving nearly all applicants who clear a modest 0.35 score expands financial inclusion, but directly increases bad debt write-offs if the bank cannot absorb the unpaid principal.

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**Question 8: Demographic Parity vs. Equalized Odds**

For commercial bank lending on the German Credit dataset, I recommend prioritizing Equalized Odds (EO) over Demographic Parity (DP). A bank loan is a binding financial contract that requires repayment; therefore, individual creditworthiness—the genuine ability and willingness to repay debt—must serve as the guiding standard for approval decisions.

Our threshold analysis in Part 2.4 demonstrates that Demographic Parity and Equalized Odds measure completely different concepts. At threshold t = 0.35, the model temporarily satisfies both metrics (DP = 1.0184, EO = 1.0080, Predictive Parity = 1.0256, accuracy = 75.00%). However, as the bank tightens approval standards toward standard industry levels like t = 0.55, DP falls to 0.8878, while EO holds at 0.8750. In our data, younger applicants naturally have shorter job tenures and smaller savings balances, leading to an empirical approval rate of 66.67% at t = 0.55 compared to 75.10% for older applicants.

Demographic Parity requires the bank to approve equal percentages of applicants across groups, regardless of whether they can afford repayment. Enforcing strict DP would force the lender to lower approval standards for younger applicants who are likely to default, setting them up for financial distress and damaging their future credit standing. In contrast, EO requires equal accuracy: it ensures that creditworthy young applicants have the same True Positive Rate as creditworthy older applicants, while keeping False Positive Rates balanced.

By prioritizing EO, we intentionally sacrifice DP under tighter lending conditions. This trade-off is ethically defensible because it treats every qualified applicant fairly based on actual merit, avoiding both discriminatory exclusion and irresponsible over-lending that threatens bank solvency.

## PART 4: DEPLOYMENT GOVERNANCE REFLECTION

> *"If a policymaker asked you: 'Just give me one number: should we deploy this model or not?', what would you say, and what is the limitation of that 'one number' answer that you discovered yourself through your analysis?"*

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### 1. Direct Recommendation

I recommend conditional deployment with mandatory human review for borderline applicants and separate subgroup evaluations, because while the model achieves 74.67% test accuracy, it conceals an 8.67% high-confidence error rate and severe differences across demographic groups.

### 2. Why a Single Number Fails

First, a single overall accuracy number hides deep performance differences between demographic groups. In our Simpson's Paradox audit, the model achieved stable overall accuracy, yet underlying feature weights completely flipped across genders. For example, how long an applicant lived at their residence reversed from a slightly negative coefficient in the full sample (beta = -0.0194) to a positive benefit for men (beta = +0.0304). Similarly, loan duration strongly predicted defaults for male borrowers ($p = 0.002$) but had no statistical relationship for female borrowers ($p = 0.509$). A single accuracy score gives a false sense of security while applying a one-size-fits-all formula that misjudges distinct groups.

Second, choosing a decision cutoff is a policy value judgment rather than a neutral mathematical fact. An overall score assumes all errors carry equal costs, but changing the threshold shifts who bears the financial burden. Lowering the cutoff to $t = 0.35$ maximizes test accuracy (75.00%) and achieves balanced approval rates between young and older borrowers (Demographic Parity [DP] = 1.0184), but exposes the bank to an elevated young false positive rate of 72.73%. Conversely, setting a strict cutoff of $t = 0.55$ shields the bank from unpaid debt but rejects 33.33% of creditworthy younger borrowers. A single metric cannot tell a policymaker which group should absorb risk.

Third, different fairness standards directly clash with one another. When demographic groups have different baseline repayment rates, mathematical theorems prove that no model can satisfy Demographic Parity, Equalized Odds (EO), and Predictive Parity at the same time. At threshold $t = 0.35$, the model balances approval rates across age groups, but as the bar is raised to $t = 0.55$, DP falls to 0.8878. A single approval number hides the reality that helping one group under one definition inevitably harms another under a different definition.

Finally, an aggregate score conceals how models fail when key variables are omitted. When we removed checking account records, savings coefficients artificially doubled (+107.30% shift), proving that clean top-line numbers can hide borrowed correlations and blind-spot failures.

### 3. Grounding in Empirical Evidence

Our stress-tests anchor these limitations in exact numbers: the 27.33% decision disagreement between linear and logistic models (82 mismatched cases); the 26 high-confidence test mistakes (8.67% error rate), where 45.45% of false positives lacked checking accounts; parameter swings reaching +260.27% under omitted variable testing; and a 1.03 trade-off cost ratio between DP and EO during threshold sweeps.

### 4. Closing Meta-Lesson

Responsible deployment of artificial intelligence cannot be answered with a single pass or fail score. Real-world deployment requires clear transparency about unavoidable trade-offs between bank solvency and social equity,

continuous subgroup auditing, and accessible human appeal processes for individuals flagged by automated systems.

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> **Verification & Reproducibility Guarantee:** All empirical numbers, regression coefficients, sample counts, and fairness metrics reported in this document were verified directly from the German Credit dataset by running the accompanying Jupyter notebook ( `german_credit_audit.ipynb` ) with random seed `343` .